

What is a String?

- A string is a sequence of characters.
- It is written inside single quotes (' '), double quotes (" "), or triple quotes (' ' ' ' ' ').
- Strings are **immutable**, meaning they cannot be changed after they are created

Built-in Functions

len() -Returns the total number of characters in a string.

Indexing

- Used to access a single character from a string.
- Positive indexing starts from **0**.
- Negative indexing starts from **-1** from the end.

Slicing

- Used to extract a part of a string.
- It returns a new string containing the selected characters.

upper() -Converts all characters to uppercase.

lower() -Converts all characters to lowercase.

split()

- Splits a string into a list based on a separator.
- By default, it splits using spaces.

count()

- Counts how many times a character or word appears in a string.

find()

- Returns the index of the first occurrence of a character or word.
- Returns **-1** if it is not found.

Print Formatting

Normal Print-Displays values using commas.

.format() - Inserts values into placeholders { } inside a string.

f-String

- Uses **f** before the string.
- Variables are placed inside { }.
- It is the easiest and most commonly used formatting method.

String Checking Functions

isspace() -Returns **True** if the string contains only spaces.

istitle() -Returns **True** if every word starts with a capital letter.

isupper() -Returns **True** if all alphabetic characters are uppercase.

endswith() -Checks whether a string ends with a specific character or word.

Lists

What is a List?

- A list is a collection of multiple items.
- Lists are **ordered**.
- Lists are **mutable**, meaning they can be modified.
- Lists allow duplicate values.
- Lists can store different data types together.

Difference Between String and List

String	List
Immutable	Mutable
Stores characters	Stores multiple values
Uses quotes	Uses square brackets []
Cannot modify characters	Can modify elements

Pop()

- Removes an element from a list.
- By default, it removes the last element.
- It also returns the removed value.

sort()

- Arranges list elements in ascending order by default.
- Can also sort in descending order.

reverse() - Reverses the order of elements in a list.

Matrix

- A matrix is a **two-dimensional list**.
- It contains rows and columns.
- Each row is itself a list.

Nested List

- A nested list is a list that contains one or more lists as its elements.
- Used to represent structured or grouped data.

List Comprehension

- A short and efficient way to create a new list.
- Combines loops and optional conditions into a single line.
- Makes code shorter and easier to read.

Tuples

What is a Tuple?

- A tuple is a collection of multiple items.
- Tuples are **ordered**.
- Tuples are **immutable**.
- Tuples allow duplicate values.
- Tuples are generally faster than lists because they cannot be modified.

4. Sets

What is a Set?

- A set is a collection of unique values.
- Sets are **unordered**.
- Sets do not allow duplicate values.
- Sets are mutable, but their elements must be immutable.

5. Dictionaries

What is a Dictionary?

- A dictionary stores data as **key-value pairs**.
- Every key must be unique.
- Values can be duplicated.
- Dictionaries are mutable.
- Dictionaries provide fast data lookup using keys.

Dictionary Comprehension

- A short way to create dictionaries.
- Used to generate key-value pairs in a single line.
- Improves readability and reduces code length.

Shallow Copy

- Creates a copy of the outer object only.
- Nested objects are **shared** between the original and copied object.
- Changes to nested objects affect both copies

Deep Copy

- Creates a completely independent copy.
 - Both the outer object and all nested objects are copied.
 - Changes made to the copied object do not affect the original object.
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Quick Revision

Strings

- Sequence of characters
- Immutable
- Supports indexing and slicing

String Functions

- `len()` → Length
- `upper()` → Uppercase
- `lower()` → Lowercase
- `split()` → Split into list
- `count()` → Count occurrences
- `find()` → Find position
- `isspace()` → Only spaces
- `istitle()` → Title case check
- `isupper()` → Uppercase check
- `endswith()` → Ending check

Lists

- Ordered
- Mutable
- Allows duplicates

Tuples

- Ordered
- Immutable
- Allows duplicates

Sets

- Unordered
- No duplicates

Dictionaries

- Key-value pairs
- Mutable
- Keys are unique

Comprehension

- Short way to create lists or dictionaries

Copy

- **Shallow Copy** → Nested objects are shared.
- **Deep Copy** → Everything is copied independently.